

Answer: B

- 27** Since the cross-sectional area is constant throughout, the number of free electrons passing through the cross-sectional area per unit time at both the copper and semiconductor material should be the same (i.e. constant current).

Total number of electrons passing through cross-sectional area per unit time
$$= (8.6 \times 10^{28} \text{ m}^{-3})(0.58 \times 10^{-3} \text{ m s}^{-1}) = 4.3 \times 10^{21} \text{ m}^{-3} \times v_e$$

$$v_e = 1.16 \times 10^4 \text{ m s}^{-1}$$

Answer: C